COMPETENCY STANDARD

FOR

WELDING (1G, 2G & 3G)

(LIGHT ENGINEERING SECTOR)

Skills for Industry Competitiveness and Innovation Program (SICIP)
Finance Division, Ministry of Finance

COPYRIGHT

This document is available at:
Skills for Industry Competitiveness and Innovation Program (SICIP)
Finance Division, Ministry of Finance,
Probashi Kallyan Bhaban (Level-15 & 16)
71-72 Old Elephant Road, Eskaton Garden
Dhaka-1000
Phone: +8802 55138753-55, Fax: +88 02 55138752
Website: https://sicip.gov.bd/

INTRODUCTION:

The **Skills for Industry Competitiveness and Innovation Program (SICIP)** has the overall objective of developing a skilled workforce adept at handling new technologies, especially for emerging industries in Bangladesh. It will expand skills training and strengthen the development of the training ecosystem to address the skills requirements of the SICIP-selected industry sectors. The program aims to (i) increase the technology-oriented skilled workforce across emerging and priority sectors, (ii) promote inclusive skilling and upskilling opportunities for women and socially disadvantaged groups, (iii) incentivize industry-university partnerships to nurture innovation capacity and improve industry competitiveness, and (iv) foster skills for climate-resilient manufacturing processes and green technologies. The program is expected to benefit about 220,000 new and existing workers over a 6-year implementation period from 2024-2029.

The SICIP Program has, therefore, taken the initiative to enhance the employability and productivity of trainees by implementing market-responsive and job-focused training programs through public and private training providers. This will require the development of competency standards for each of the occupations/trades which will provide a structured framework in the learning process to guide training providers, ensure consistent training quality, and create an alignment between the skills provided by the training institutes and the needs of the industry.

<u>The Competency Standard also suggests integration of YouTube or similar platforms or downloaded clips into classroom practice to ensure simulated creation of the contents so that learners are exposed to visual demonstrations before classroom instruction or practical session, which aligns with modern learning preference and supports flipped classroom models.</u>

This competency standard is therefore developed to improve skills following the job roles and skill sets of the occupation and ensure that the required skills are aligned with industry requirements.

The document details the format, sequencing, wording, and layout of the Competency Standard for an occupation which comprises Units of Competence and

OVERVIEW:

A **Competency Standard** is a written specification of the knowledge, skills, and attitudes required for the performance of a job or occupation or trade corresponding to the standard of performance required in the workplace.

Competency standard:

- provides a consistent and reliable set of components for training, recognizing, and assessing people's skills, and may also have optional support materials.
- enables industry-recognized qualifications to be awarded through direct assessment of workplace competencies
- encourages the development and delivery of flexible training that suits individual and industry requirements
- encourages learning and assessment in a work-related environment which leads to verifiable workplace outcomes.

Competency Standard has been developed by a working group comprised of occupation-specific experts from the industry/institution and relevant consultants of SICIP.

Competency Standards describe the skills, knowledge, and attitude needed to perform effectively in the workplace. Competency Standards acknowledge that people can achieve vocational and technical competency in many ways by emphasizing what the learner can do, not how or where they learned to do it.

With Competency Standards, assessment and training may be conducted at the workplace, at training organization, during regular work, or through work experience, work placement, work simulation or any combination of these.

A Unit of Competency describes a distinct work activity that would normally be undertaken by one person in accordance with industry standards.

Units of Competency are documented in a standard format that comprises:

- Reference to Industry Sector, Occupational Title and Occupational Description
- Unit code
- Unit title
- Unit descriptor
- Elements and performance criteria
- Variables and range statement
- Evidence guides

Together all the parts of a Unit of Competency:

- Describe a work activity
- Guide the assessor in determining whether the candidate is competent.

Identification and validation of units of competency and elements for each occupation were made by experts from Construction industry in consultative workshop.

The ensuing sections of this document comprise a description of the respective occupation with all the key components of a Unit of Competence:

- An overview of all Units of Competence for the occupation and their corresponding duration required for completion of training.

- The Competency Standards that include the Unit of Competency, Unit Descriptor, Elements and Performance Criteria, Range of Variables, Curricular Content Guide, and Assessment Evidence Guide.

Units & Elements at a Glance:

Generic Competencies (30 Hrs.)

Code	Unit of Competency	Elements of Competency	Duration (Hours)
SICIP-LE-WEL-01-G	Apply occupational health and safety (OHS) practice in the workplace	1. Identify OHS policies and procedures. 2. Apply personal health and safety practices. 3. Report hazards and risks 4. Respond to emergencies.	10
SICIP-LE-WEL-02-G	Perform Computations Using Basic Mathematical Concepts	1. Identify calculation requirements in the workplace 2. Select appropriate mathematical methods 3. Perform calculations	10
SICIP-LE-WEL-03-G	Interpret Technical Drawings and Manuals	1. Select technical drawing 2. Interpret technical drawings. 3. Interpret operation & maintenance manuals	10
		Total Hour	**30**

Sector Specific Competencies (30 Hrs.)

Code	Unit of Competency	Elements of Competency	Duration (Hours)
SICIP-LE-WEL-01-S	Apply Green Practices in Light Engineering Sector	1. Interpret green concepts 2. Minimize resource use in the workplace 3. Implement waste management practices	10
SICIP-LE-WEL-02-S	Use Hand and Power Tools	1. Identify and inspect hand and power tools 2. Use hand tools properly and safely 3. Operate power tools properly and safely 4. Clean and maintain hand tools and power tools	10
SICIP-LE-WEL-03-S	Carry Out Precision Checks and Measurements	1. Select the job to be checked and measured 2. Select measuring and checking tool/instrument 3. Perform measurements 4. Check and record measurement 5. Clean, maintain and store measuring instruments.	10
		Total Hour	**30**

Occupation Specific Competencies (300 Hrs.)

Code	Unit of Competency	Elements of Competency	Duration (hours)
SICIP-LE-WEL-01-O	Apply Welding Process and Techniques	1 Understand welding process and techniques 2 Identify materials used in welding 3 Recognize common welding defects 4 Interpret quality control standards in welding	20
SICIP-LE-WEL-02-O	Prepare Materials and Equipment for Welding	1. Select base materials and consumables 2. Arrange welding tools 3. Identify welding joints 4. Set up welding machines 5. Adjust welding parameters for each welding position	20
SICIP-LE-WEL-03-O	Carry Out Shielded Metal Arc Welding	1. Identify and prepare work requirements 2. Select welding job, equipment and job holding devices 3. Perform welding in different positions 4. Clean/maintain the workplace	80
SICIP-LE-WEL-04-O	Perform Gas Welding	1. Prepare for gas welding 2. Perform gas welding 3. Carry out gas cutting 4. Clean and store tools and equipment	60
SICIP-LE-WEL-05-O	Carry Out Gas Metal Arc Welding	1. Identify and prepare work requirements 2. Select welding job, equipment and job holding devices 3. Perform GMAW or MIG welding 4. Clean/maintain the workplace	60
SICIP-LE-WEL-06-O	Carry Out Gas Tungsten Arc Welding	1. Identify and prepare work requirements 2. Select welding job, equipment and job holding devices 3. Perform GTAW or TIG welding 4. Clean/maintain the workplace	60
		Total Hours	**300**

COMPETENCY STANDARD: WELDING (1G, 2G & 3G)

Generic Competencies

Unit of Competency: **APPLY OCCUPATIONAL HEALTH AND SAFETY (OHS) PRACTICE IN THE WORKPLACE**	**Nominal Duration:** 10 hrs.	**Unit Code:** SICIP-LE-WEL-01-G
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to apply occupational health and safety (OHS) practices in the workplace. It specifically includes the tasks of identifying OHS policies and procedures, applying personal health and safety practices, reporting hazards and risks and responding to emergencies.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify OHS policies and procedures	1.1 **<u>OHS policies</u>** and safe operating procedures are interpreted 1.2 Safety signs and symbols are identified and followed. 1.3 Response, evacuation procedures and other contingency measures are interpreted correctly.
2. Apply personal health and safety practices	1.4 OHS policies and procedures are interpreted in the workplace including **<u>personal protective equipment (PPE).</u>** 1.5 Common health issues are recognized. 1.6 Common safety issues are identified and applied.
3. Report hazards and risks	1.1 Hazards and risks are identified. 1.2 Hazards and risks assessment and controls are interpreted 1.3 Hazards and risk assessment are reported
4. Respond to emergencies	1.1 Respond to alarms and warning devices. 1.2 **<u>Emergency response plans and procedures</u>** are responded to. 1.3 **<u>First aid procedures</u>** during emergency situations are identified.

Range of variables:

Variables	**Range** (may include but not limited to)
1. OHS policies	1.1 Organizational OHS polices 1.2 International OHS requirements 1.3 Fire safety rules and regulations
2. Personal protective equipment	2.1 Safety glasses 2.2 Ear plugs 2.3 Gloves 2.4 Apron 2.5 Helmet 2.6 Mask 2.7 Safety shoes
3. Emergency response plans and procedures	3.1 Firefighting procedures 3.2 Earthquake response procedures 3.3 Emergency response plans and procedures 3.4 Medical and first aid
4. First aid procedure	4.1 Washing of open wound 4.2 Washing chemically infected area 4.3 Applying bandage 4.4 Taking appropriate medicine

Curricular Content Guide

1. Underpinning knowledge	1.1 Workplace OHS policies and procedures 1.2 Work safety procedures 1.3 Emergency response procedures: 1.3.1 Firefighting 1.3.2 Earthquake response 1.3.3 Accident response 1.4 Types of hazards (biological, chemical and physical) and their effects 1.5 OHS awareness 1.6 Personal protective equipment (PPE) 1.7 Malfunctions and resolutions
2. Underpinning skill	1.8 Identifying OHS policies and procedures 1.9 Applying personal health and safety practices 1.10 Reporting hazards and risks 1.11 Responding to emergencies
3. Underpinning Attitudes	1.12 Commitment to occupational health and safety 1.13 Promptness in carrying out activities 1.14 Sincere and honest to duties 1.15 Environmental concerns 1.16 Eagerness to learn 1.17 Tidiness and timeliness 1.18 Respect for rights of peers and seniors in the workplace 1.19 Communication with peers, subordinates and seniors in the workplace

4. Resource implications	The following resources must be provided: 1.20 Workplace (simulated or actual) 1.21 Workplace documents, signs and symbols 1.22 Codes of conduct 1.23 Projector 1.24 Learning manual 1.25 Tools, equipment and facilities appropriate to the process or activities. 1.26 Materials are relevant to the proposed activity.

Assessment Evidence Guide

1. Critical aspects of competency	Assessment required evidence that the candidate: 1.1 identified OHS policies and procedures 1.2 applied personal health and safety practices (including PPE) 1.3 reported hazards and risks 1.4 responded to emergencies.
2. Methods of assessment	Methods of assessment may include but not limited to: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: **PERFORM COMPUTATIONS USING BASIC MATHEMATICAL CONCEPTS**	Nominal Duration: 10 hrs.	Unit Code: SICIP-LE-WEL-02-G
Unit Descriptor: This unit of competency requires knowledge, skills and attitude to perform computations using basic mathematical concepts in workplace. It specifically includes the tasks of identifying calculation requirements in the workplace, selecting appropriate mathematical methods, and performing calculations.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify calculation requirements in the workplace	1.1 **Calculation requirements** are identified based on the tasks and specifications. 1.2 Calculation requirements are understood. 1.3 **Workplace information** is interpreted to determine the necessary calculations. 1.4 Relevant workplace documents are identified to gather necessary data for calculations.
2. Select appropriate mathematical methods	2.1 **Appropriate calculation method** is identified based on the specific requirements of the task. 2.2 Units of measurement are considered when selecting the method for conversion. 2.3 Selected calculation method is applied to verify its effectiveness before proceeding with the final calculation.
3. Perform calculations	3.1 Appropriate **tools and instruments** are identified based on the requirements of the calculation. 3.2 Tools and instruments are checked for accuracy and calibration. 3.3 Computation is made using the identified tools and instruments. 3.4 Final result is checked to verify the correctness of the calculation before reporting.

Range of variables:

Variables	**Range** (may include but not limited to)
1. Calculation requirements.	1.1 Area 1.2 Volume 1.3 Measurements: 1.3.1 Length

	1.3.2 Width 1.3.3 Height 1.3.4 Thickness 1.3.5 Diameter
2. Workplace information	2.1 Drawing 2.2 Part drawing 2.3 Verbal instructions 2.4 Job order and Job sheet
3. Appropriate calculation method	3.1 Addition 3.2 Subtraction 3.3 Division and Multiplication 3.4 Conversion of measurement 3.5 Percentage and ratio calculation 3.6 Geometric Formulas (Square, Rectangle, Circle, Sphere, etc.)
4. Tools and instruments	4.1 Calculator 4.2 Computer

Curricular Content Guide

1. Underpinning knowledge	1.1 Numerical concept, identification of calculation requirements 1.2 Basic mathematical methods, such as addition, subtraction, multiplication, division, percentage, conversion, ratio and basic geometry formulas 1.3 Mathematical language, symbols 1.4 Measuring units
2. Underpinning skill	2.1 Computing basic mathematics 2.2 Practicing conversion of measurements 2.3 Calculating areas, volume, etc. using formulas
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4. Resource implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Standard operating procedure 4.3 Workplace documents, signs and symbols 4.4 Codes of conduct

	4.5 Projector 4.6 Learning manual 4.7 Tools, equipment and facilities appropriate to the process or activities. 4.8 Materials are relevant to the proposed activity.

Assessment Evidence Guide

1. Critical aspects of competency	Assessment required evidence that the candidate: 1.1 identified calculation requirements from workplace information. 1.2 selected appropriate method to carry out the calculation requirements. 1.3 completed calculations using appropriate tools/instruments.
2. Methods of assessment	Methods of assessment may include but not limited to: 1.4 Written test 1.5 Practical Demonstration 1.6 Oral Questioning 1.7 Portfolio (Optional)
3. Context of assessment	1.8 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 1.9 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: **INTERPRET TECHNICAL DRAWINGS AND MANUALS**	**Nominal Duration:** 10 hrs.	**Unit Code:** SICIP-LE-WEL-03-G
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to translate technical drawings and plans. It specifically includes the tasks of selecting technical drawing, interpreting operation and maintenance manuals.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Select technical drawing	1.1 Appropriate **technical drawing** is selected based on the specifications and requirements of the job. 1.2 Selected drawing is checked to ensure that it accurately reflects the job requirements.
2. Interpret technical drawings.	2.1 Drawing components, assemblies are identified 2.2 Dimensions are identified according to job requirement 2.3 Clearances/tolerances are checked in accordance with workplace standard 2.4 **Instructions** are identified and followed accurately. 2.5 Material **specifications** are interpreted 2.6 Symbols in drawing are interpreted.
3. Interpret operation & maintenance manuals	3.1 Operation and maintenance manuals are collected and interpreted 3.2 Operation and maintenance manuals are followed.

Range of variables:

Variables	**Range** (may include but not limited to)
1. Technical drawing	1.1 Sketches 1.2 Manuals
2. Instructions	2.1 Note 2.2 Instruction 2.3 Special instruction 2.4 Precaution
3. Specifications	3.1 Product specifications 3.2 Method specifications 3.3 Material specifications

Curricular Content Guide

1. Underpinning knowledge	1.1 Technical drawing interpretation 1.2 Sequence of drawing 1.3 Methods of checking and applying drawing for work

	1.4 Drawing selection and checking method to ensure conformity to the job requirements. 1.5 Drawing components, assemblies 1.6 Identification of dimensions according to job requirement 1.7 Procedure of checking clearances/tolerances 1.8 Work instructions 1.9 Material specifications 1.10 Drawing symbols interpretation 1.11 Use of operation and maintenance manuals
2. Underpinning skill	2.1 Practicing workplace safety 2.2 Interpreting drawing, following operation and maintenance manuals, 2.3 Performing jobs in accordance with the drawing 2.4 Performing calculation as per drawing 2.5 Selecting and checking of drawing to ensure conformity to the job requirements. 2.6 Identifying drawing components and assemblies 2.7 Identifying dimensions according to job requirement 2.8 Checking clearances/tolerances in accordance with workplace standard 2.9 Following operation and maintenance manuals
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Eagerness to learn 3.5 Tidiness and timeliness 3.6 Environmental concerns 3.7 Respect for rights of peers and seniors at workplace 3.8 Communication with peers and seniors at workplace
4. Resource implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Workplace documents, Projector 4.3 Learning manual 4.4 Tools, equipment and facilities appropriate to the process or activities. 4.5 Materials are relevant to the proposed activity.

Assessment Evidence Guide

1. Critical aspects of competency	Assessment required evidence that the candidate: 1.1 identified dimension according to job requirement 1.2 maintained clearances and tolerances according to workplace requirement. 1.3 interpreted technical drawing 1.4 interpreted operation & maintenance manuals
2. Methods of assessment	Methods of assessment may include but not limited to:

	2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

The Sector Specific Competencies

Unit of Competency: **APPLY GREEN PRACTICES IN LIGHT ENGINEERING SECTOR**	**Nominal Duration:** 10 hrs.	**Unit Code:** SICIP-LE-WEL-02-S
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to apply green practices in shipbuilding sector. It specifically includes the tasks of interpreting green concepts, minimizing resources use in the workplace and implementing waste management practices.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Interpret green concepts	1.1 **<u>Basic concepts of green</u>** are understood. 1.2 Energy efficiency principles are interpreted. 1.3 **<u>Sustainable materials</u>** are identified. 1.4 Pollution control methods are understood. 1.5 **<u>Environmental certifications and standards</u>** are identified. 1.6 **<u>Green supply chains</u>** are interpreted to minimize environmental impact.
2. Minimize resources use in the workplace	2.1 Resource consumption is identified in the light engineering sector. 2.2 **<u>Energy-efficient equipment</u>** is identified and maintained. 2.3 Materials are sourced ensuring that sustainable and recyclable materials are prioritized. 2.4 Water usage is minimized through efficient management. 2.5 Production processes are optimized. 2.6 **<u>Green technologies and innovations</u>** are implemented in the workplace.
3. Implement waste management practices	3.1 **<u>Waste types</u>** identified in the light engineering sector. 3.2 Waste segregation practices are interpreted. 3.3 Recycling programs are established through external recycling facilities. 3.4 **<u>Waste reduction strategies</u>** are understood. 3.5 Safe disposal practices are described. 3.6 **<u>Waste-to-energy solutions</u>** are explored to reduce landfill contributions.

Range of Variables

Variable	Range

	May include but not limited to:
1. Basic concepts of green	1.1 Principles of Green Practices 1.2 Sources of Environmental Impacts 1.3 Resource Utilization 1.4 Sustainable Alternatives 1.5 Waste Management Practices 1.6 Green Habits at Work 1.7 Compliance and Regulations
2. Sustainable materials	2.1 Recycled Metals 2.2 Biodegradable Lubricants and Fluids 2.3 Composite Materials with Low Environmental Impact 2.4 Low-VOC Paints and Coatings 2.5 Eco-friendly Packaging Materials 2.6 Durable and Repairable Components
3. Environmental certifications and standards	3.1 ISO 14001: Environmental Management System (EMS) 3.2 LEED Certification (Leadership in Energy and Environmental Design) 3.3 RoHS (Restriction of Hazardous Substances) 3.4 REACH (Registration, Evaluation, Authorization and Restriction of Chemicals) 3.5 Energy Star Certification 3.6 Eco-Labeling Standards 3.7 Green Building Codes & Local Environmental Compliance
4. Green supply chains	4.1 Environmentally Responsible Sourcing 4.2 Eco-Friendly Transportation and Logistics 4.3 Supplier Environmental Compliance 4.4 Sustainable Packaging and Distribution
5. Energy-efficient equipment	5.1 High-Efficiency Electric Motors (IE2/IE3/IE4 Rated) 5.2 Variable Frequency Drives (VFDs) 5.3 Energy-Efficient LED Lighting Systems 5.4 Energy-Star Rated Industrial Equipment 5.5 Heat Recovery and Insulation Systems
6. Green technologies and innovations	6.1 Renewable Energy Integration 6.2 Additive Manufacturing (3D Printing) 6.3 Smart Manufacturing Systems (Industry 4.0) 6.4 Waste-to-Energy Technologies 6.5 Low-Emission Welding and Cutting Equipment 6.6 Closed-Loop Cooling and Lubrication Systems 6.7 Eco-Innovation in Product Design
7. Waste types	7.1 Metal Waste 7.2 Plastic and Packaging Waste 7.3 Hazardous Waste 7.4 Electronic Waste (E-Waste)

	7.5 Dust and Particulate Waste 7.6 Wood and Pallet Waste
8. Waste reduction strategies	8.1 5S Workplace Organization 8.2 Lean Manufacturing Practices 8.3 Material Optimization and Nesting Techniques 8.4 Reusing and Repurposing Waste
9. Waste-to-energy solutions	9.1 Thermal Conversion (Incineration with Energy Recovery) 9.2 Gasification and Pyrolysis 9.3 Biomass Boilers Using Organic Waste 9.4 Use of Waste Oils as Alternative Fuel 9.5 Waste Heat Recovery Systems (WHRS) 9.6 On-Site Micro Energy Generation Units

Curricular Content Guide

1. Underpinning Knowledge	1.1 Principles of pollution. 1.2 Pollution control technologies 1.3 Legal and environmental frameworks: overview of national environmental policies and local environmental compliance regulations. 1.4 Importance of ISO 14001 and other EMS (Environmental Management System) standards. 1.5 Classification of sustainable, recyclable, and hazardous materials used in production. 1.6 Strategies for minimizing water and energy use during material processing. 1.7 Waste hierarchy: reduce, reuse, recycle, recover, and disposal (5Rs). 1.8 Industrial waste-to-energy concepts.
2. Underpinning Skills	2.1 Measuring and analyzing energy and water consumption using meters. 2.2 Identifying and operating energy-efficient machinery. 2.3 Monitoring and logging resource consumption data. 2.4 Adjusting machine settings or workflow layouts. 2.5 Applying green inventory practices. 2.6 Segregating waste effectively into recyclable, hazardous, and general categories. 2.7 Implementing basic pollution control activities.
3 Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace

	3.8 Communication with peers, subordinates and seniors in the workplace
4 Resource Implications	4.1 Workplace (simulated or actual) 4.2 Projector 4.3 Stationery 4.4 Learning manual

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 interpreted green concepts 1.2 minimized resources use in the workplace 1.3 implementing waste management practices
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: **USE HAND AND POWER TOOLS**	**Nominal Duration:** 10 hrs.	**Unit Code:** SICIP-LE-WEL-02-S
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to use hand and power tools in the workplace. It specifically includes the task of identifying and inspecting hand and power tools, using hand tools properly and safely, operating power tools properly and safely and cleaning and maintaining hand and power tools.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **<u>bold and underlined</u>** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify and inspect hand and power tools	1.1 Appropriate hand and power tools are identified. 1.2 Application of hand and power tools is recognized. 1.3 Usability of hand and power tools is checked and verified.
2. Use hand tools properly and safely	2.1 Appropriate **<u>hand tools</u>** are selected. 2.2 Safety precautions are ensured before using hand tools. 2.3 Unsafe or faulty hand tools are identified and marked for repair. 2.4 **<u>Measuring tools</u>** are checked and calibrated before use. 2.5 Use hand tools properly and safely to perform work activity.
3. Operate power tools properly and safely	3.1. Appropriate **<u>power tools</u>** are selected. 3.2. Power supply outlet and electrical cord are inspected and confirmed safe for use in accordance with established workplace safety requirements. 3.3. Safety precautions are ensured before using power tools in accordance with manufacturer's operating specification. 3.4. Proper sequence of operation is maintained using power tools. 3.5. Unsafe or faulty power tools are identified and marked for repair. 3.6. Power tools are operated properly and safely to perform bench work.
4. Clean and maintain hand and power tools	4.1. Dust and foreign matter are removed from hand and power tools in accordance to workplace standards. 4.2. Condition of hand and power tools is checked after use and reported. 4.3. Appropriate lubricant is applied after use and prior to storage.

	4.4. Defective hand and power tools are inspected and repaired or replaced. 4.5. Hand and power tools are stored and secured in accordance with workplace requirements.

Range of Variables

Variable	Range May include but not limited to:
1. Hand tools	1.1 Hammer 1.2 Bench vice 1.3 Files 1.4 Wrenches 1.5 Pliers 1.6 Scriber 1.7 Scraper Screw 1.8 Surface plate 1.9 Gauge 1.10 Tap sets 1.11 Die sets 1.12 Hacksaw 1.13 Spanners 1.14 Vice grip 1.15 Wire cutters 1.16 Clamps 1.17 Jacks
2. Power tools	2.1 Drills 2.2 Rivet gun 2.3 Grinders 2.4 Pneumatic wrenches 2.5 Press machine 2.6 Cutting 2.7 Saws 2.8 Soldering iron
3. Measuring tools	3.1 Micrometer 3.2 Megger 3.3 Measuring tape 3.4 Hose level 3.5 Water level 3.6 Calipers 3.7 Steel rule 3.8 Meter rule 3.9 Spirit level 3.10 Protractor 3.11 Tri-square 3.12 Gauges

Curricular Content Guide

1. Underpinning Knowledge	1.1 Information on hand and power tools, their functions and use 1.2 Procedures for safely using hand and power tools
2. Underpinning Skills	2.1 Identifying hand, power and measuring tools 2.2 Following safety precautions when using hand, power and measuring tools 2.3 Using hand and measuring tools correctly and safely in accordance with manufacturer's operating specification 2.4 Operating power tools correctly and safely in accordance with manufacturer's operating specification 2.5 Cleaning and maintaining hand and power tools after use 2.6 Applying appropriate lubricant on hand and power tools after use and prior to storing
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	4.1 Workplace (simulated or actual) 4.2 Standard operating procedure 4.3 Workplace documents, signs and symbols 4.4 Codes of conduct 4.5 Projector 4.6 Learning manual 4.7 Tools, equipment and facilities appropriate to the process or activities. 4.8 Materials are relevant to the proposed activity.

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 identified and selected appropriate hand and power tools for work to be performed 1.2 identified and used measuring and testing tools appropriate to work activity 1.3 followed safety precautions when using hand and power tools 1.4 operated power tools safely and pursuant to

	manufacturer's operating specification 1.5 performed cleaning and maintenance of hand and power tools after use and prior to storing
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: **CARRY OUT PRECISION CHECKS AND MEASUREMENTS**	**Nominal Duration:** 10 hrs.	**Unit Code:** SICIP-SBD-WF-03-S
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to carry out precision checks and measurements. It specifically includes the tasks of selecting the job to be measured, selecting measuring and checking tools/instrument, performing measurements, checking & recording measurements and cleaning, maintaining and storing measuring instruments.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Select the job to be checked and measured	1.1 Job is selected for measuring and checking 1.2 Required **dimensional measurement** is determined in accordance with drawing/plan 1.3 Required **physical conditions** are identified in accordance with drawing/plan 1.4 Required **geometrical dimensions** are identified in accordance with drawing/plan 1.5 Job drawing is used to select the measuring instruments.
2. Select measuring and checking tool/instrument	2.1 Appropriate measuring instruments is selected in accordance with job requirement. 2.2 **Direct** and **indirect measuring instruments** and **checking instrument** are identified 2.3 Applications of measuring device is determined. 2.4 Usability and accuracy of measuring device is checked and verified. 2.5 Measuring device is prepared for measurement. 2.6 Fits, Tolerance, clearance and limits are identified according to job requirements.
3. Perform measurements	3.1 Measurements are obtained using appropriate measuring instrument. 3.2 **Systems of measurements** are identified and converted where necessary. 3.3 Measurement is kept accurately in accordance to specification 3.4 Measurement is checked against job requirement 3.5 Physical conditions are checked in accordance with job requirements 3.6 Geometrical dimensions are checked in accordance with job specifications
4. Check and record measurement	4.1 Measurements are checked and recorded in accordance with workplace procedure

	4.2 Measurement is interpreted, recorded and communicated to authority
5. Clean, maintain and store measuring instruments.	5.1 Dust and dirt are removed from the measuring instruments 5.2 Condition of measuring instruments are checked 5.3 Appropriate lubricant is applied after use and prior to storage 5.4 Measuring instruments are checked and calibrated 5.5 Measuring instruments are stored in accordance with workplace procedure.

Range of Variables

Variable	**Range** May include but not limited to:
1. Dimensional measurement	1.1 Length 1.2 Width 1.3 Depth 1.4 Diameter 1.5 Radius 1.6 Height
2. Physical condition	2.1 Roughness 2.2 Color 2.3 Smoothness 2.4 Surface finish 2.5 Flatness
3. Geometrical dimension	3.1 Angle 3.2 Circle 3.3 Triangle 3.4 Square 3.5 Rectangle 3.6 Hexagon shape 3.7 Pentagon shape
4. Direct measuring instruments	4.1 Set squares 4.2 Dial indicators 4.3 Steel tape 4.4 Steel rule 4.5 Meter rule 4.6 Calculator 4.7 Vernier slide caliper 4.8 Digital Vernier slide caliper 4.9 Micrometer (inch/millimeter) 4.10 Digital micrometer 4.11 Vernier bevel protractor 4.12 Sprit level 4.13 AVO meter(analogue/digital) 4.14 Thermometers

	4.15 Water meter 4.16 Gas meter 4.17 Simple protractor
5. Indirect measuring instrument	5.1 Outside caliper 5.2 Inside caliper 5.3 Bevel tri-square 5.4 Telescoping gage 5.5 Straight edge 5.6 Side bar 5.7 Trammel
6. Checking instrument.	6.1 Plug gauge 6.2 Snap gauge 6.3 Screw pitch gauge 6.4 Slip gauges 6.5 Feeler gauges 6.6 Screw pitch gauge 6.7 Slip gauge 6.8 Tri-square 6.9 Center gauge 6.10 Bevel tri-square
7. Systems of measurements	7.1 ISO standard 7.2 English system 7.3 Metric system

Curricular Content Guide

1. Underpinning Knowledge	1.1 Difference between measuring and checking 1.2 Types of measuring tools and their applications 1.3 Types of checking tools and their applications 1.4 Geometrical dimensions and tolerances 1.5 Method, procedure and techniques when taking linear Measurements 1.6 Methods, procedures and techniques when checking physical conditions of work pieces 1.7 Methods, procedures and techniques when checking geometrical dimensions of work pieces 1.8 Measurement conversion systems 1.9 Workplace record keeping procedures 1.10 Preventive maintenance for measuring and checking tools 1.11 Calibration and adjustment procedures for measuring and checking tools
2. Underpinning Skills	2.1 Determining required dimensional measurements, physical conditions and geometrical dimensions in accordance with drawing/plan and workplace specification 2.2 Measuring and checking linear and geometrical dimensions within the required tolerance in accordance

	to specification 2.3 Checking physical conditions using appropriate checking tool 2.4 Identifying and converting systems of measurements where necessary. 2.5 Recording measurements in accordance with workplace procedure 2.6 Interpreting and communicating measurement to authority 2.7 Applying appropriate lubricant on measuring and checking tools and instruments after use and prior to storage 2.8 Checking condition of measuring instruments, calibrating and storing in accordance with workplace procedure
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	4.1 Workplace (simulated or actual) 4.2 Standard operating procedure 4.3 Workplace documents, signs and symbols 4.4 Codes of conduct 4.5 Projector 4.6 Learning manual 4.7 Tools, equipment and facilities appropriate to the process or activities. 4.8 Materials are relevant to the proposed activity.

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 determined required dimensional measurements, physical conditions and geometrical dimensions in accordance with drawing/plan and workplace specification 1.2 measured and checked linear and geometrical dimensions within the required tolerance in accordance to specification 1.3 checked physical conditions using appropriate checking tool 1.4 identified and converted systems of measurements

	where necessary. 1.5 recorded measurements in accordance with workplace procedure 1.6 interpreted and communicated measurement to authority 1.7 applied appropriate lubricant on measuring and checking tools and instruments after use and prior to storage 1.8 checked condition of measuring instruments, calibrated and stored in accordance with workplace procedure
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

The Occupation Specific Competencies

Unit of Competency: APPLY WELDING PROCESS AND TECHNIQUES	Nominal Duration: 20 Hrs.	Unit Code: SICIP-LE-WEL-01-O
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to apply welding process and techniques. It specifically includes the task of understanding welding process & techniques, identifying materials used in welding, recognizing common welding defects and interpreting quality control standards in welding.		

Elements and Performance Criteria:
(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Understand welding processes and techniques	1.1 The different **welding processes** are explained based on their applications. 1.2 **Welding techniques** are understood. 1.3 **Welding symbols and terminology** are interpreted from technical drawings.
2. Identify materials used in welding	2.1 **Base materials** used in welding are identified as per industry standards. 2.2 **Consumable welding materials** are identified based on the job specifications. 2.3 Appropriate **shielding gases** for different materials are identified as per the welding process. 2.4 Welding rods and wires are recognized based on their composition.
3. Recognize common welding defects	3.1 **Common welding defects** are identified in accordance with industry standards. 3.2 Welding defects are categorized as either internal or external based on the method of inspection. 3.3 Causes of welding defects are explained based on factors. 3.4 Visual appearance of defects is recognized and interpreted according to their severity.
4. Interpret quality control standards in welding	4.1 **Welding quality control standards** are interpreted according to industry regulations and relevant codes. 4.2 Importance of welder qualification and certification is understood as per the relevant certification bodies. 4.3 Inspection criteria for weld quality are understood and followed as per established guidelines.

Range of Variables

Variable	**Range** (Includes but not limited to):
1. Welding processes	1.1 Arc welding 1.2 MIG (Metal Inert Gas) welding 1.3 TIG (Tungsten Inert Gas) welding 1.4 Stick welding (Shielded Metal Arc Welding) 1.5 Submerged Arc Welding (SAW) 1.6 Flux-Cored Arc Welding (FCAW) 1.7 Laser welding 1.8 Plasma arc welding 1.9 Oxy-fuel welding
2. Welding techniques	2.1 Butt joint welding 2.2 Fillet welding 2.3 Groove welding 2.4 Corner welding 2.5 Edge welding 2.6 T-joint welding 2.7 Plug welding 2.8 Slot welding 2.9 Spot welding 2.10 Seam welding 2.11 Butt-weld with backing strip
3. Welding symbols and terminology	3.1 **Welding symbols** 3.1.1 Arrow side 3.1.2 Other side 3.1.3 Both sides (double weld) 3.1.4 Multiple welds symbol 3.1.5 Weld leg (single and double) 3.1.6 Pitch and spacing of welds 3.2 **Terminology** 3.2.1 Root face 3.2.2 Root opening 3.2.3 Throat (of the weld) 3.2.4 Leg (of the weld) 3.2.5 Face of the weld 3.2.6 Weld metal 3.2.7 Heat-affected zone (HAZ) 3.2.8 Weld penetration 3.2.9 Back-gouging 3.2.10 Root reinforcement
4. Base materials	4.1 Steel (Carbon steel, Stainless steel, Alloy steel) 4.2 Aluminum 4.3 Copper 4.4 Titanium 4.5 Nickel alloys 4.6 Brass

	4.7 Bronze 4.8 Cast iron 4.9 Magnesium alloys 4.10 High-strength low-alloy (HSLA) steel 4.11 Inconel (nickel-based superalloys) 4.12 Zirconium
5. Shielding gases	5.1 Argon 5.2 Carbon dioxide (CO_2) 5.3 Helium 5.4 Oxygen 5.5 Nitrogen 5.6 Hydrogen 5.7 Mixtures of gases 5.8 Active gases 5.9 Inert gases
6. Consumable welding materials	6.1 Welding electrodes (for SMAW) 6.2 Filler rods (for TIG welding) 6.3 Welding wire (for MIG welding) 6.4 Flux-cored wire (for FCAW) 6.5 Submerged arc welding (SAW) wire 6.6 Brazing rods and wire 6.7 Soldering materials (e.g., flux, solder wire) 6.8 Coating materials (e.g., for electrodes or rods)
7. Common welding defects	7.1 Porosity 7.2 Cracking 7.3 Incomplete fusion 7.4 Incomplete penetration 7.5 Overlap 7.6 Undercut 7.7 Spatter 7.8 Distortion 7.9 Burn-through 7.10 Underfill 7.11 Slag inclusion 7.12 Cold laps 7.13 Hot cracks 7.14 Lamellar tearing
8. Welding quality control standards	8.1 American Welding Society (AWS) 8.2 International Organization for Standardization (ISO) 8.3 American Society of Mechanical Engineers (ASME) 8.4 European Union (EU) welding standards (EN standards) 8.5 British Standards Institution (BSI) 8.6 International Institute of Welding (IIW) 8.7 Canadian Standards Association (CSA) 8.8 ASME Boiler and Pressure Vessel Code (BPVC) 8.9 Welding Research Council (WRC)

1. Underpinning Knowledge	1.1 Identify different welding processes based on material type, thickness, and joint quality. 1.2 Control heat, speed, and electrode positioning for accurate and strong welds. 1.3 Use standardized symbols on technical drawings to indicate weld types, dimensions, and other conditions. 1.4 Select base materials based on properties like strength and melting point. 1.5 Choose filler metals and fluxes to match the base material and welding process. 1.6 Select shielding gases to protect the weld pool from contamination based on the material and process. 1.7 Identify common defects
2. Underpinning Skills	2.1 Using different welding processes based on their specific applications. 2.2 Applying appropriate welding techniques for various tasks. 2.3 Categorizing welding defects as either internal or external based on inspection methods. 2.4 Maintaining welding quality control standards according to industry regulations and relevant codes. 2.5 Following inspection criteria for weld quality as per established guidelines.
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety. 3.2 Promptness in carrying out activities. 3.3 Sincere & honest to duties. 3.4 Eagerness to learn. 3.5 Tidiness & timeliness. 3.6 Environmental concerns. 3.7 Respect for rights of peers and seniors at workplace. 3.8 Communication with peers and seniors at workplace
4 Resource Implications	The following resources must be provided: 4.1 workplace (actual or simulate) 4.2 materials relevant to proposed activities 4.3 tools, equipment and documentation 4.4 specifications or work instructions

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 understood welding principles and processes 1.2 identified materials used in welding 1.3 recognized common welding defects 1.4 interpreted quality control standards in welding
2. Methods of Assessment	2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning

	2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: **PREPARE MATERIALS AND EQUIPMENT FOR WELDING**	**Nominal Duration:** 20 Hrs**.**	**Unit Code:** SICIP-LE-WEL-02-O
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to prepare material and equipment for welding. It specifically includes the task of selecting base materials and consumables, arranging welding tools, identifying welding joints, setting up welding machines and adjusting welding parameters for each welding position.		

Elements and Performance Criteria:
(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Select base materials and consumables	1.1 Relevant **welding procedure specifications (WPS)** and **standards** are reviewed and interpreted. 1.2 **Base materials** are identified. 1.3 Welding consumables are selected based on base materials, welding position. 1.4 Welding consumables are inspected for defects.
2. Arrange welding tools	2.1 Required **welding tools** are identified. 2.2 Welding tools are inspected to ensure they are clean, functional, and free from defects. 2.3 Welding tools are arranged systematically in designated areas for easy accessibility during work. 2.4 Tools are stored securely to prevent damage and to maintain a safe working environment.
3. Identify welding joints	3.1 Safety precautions and **PPE** are used when inspecting and handling welding joints. 3.2 **Welding joint types** are identified. 3.3 Dimensions, angles, and fit-up of joints are identified.
4. Set up welding machines	4.1 Relevant welding machine manuals, specifications, and setup instructions are reviewed and understood. 4.2 The welding machine is inspected for any damage, wear, and safety compliance before setup. 4.3 Correct power supply is identified, and the welding machine is connected safely to the power source. 4.4 Required welding parameters are set according to the welding procedure specifications. 4.5 Electrodes are installed and fed correctly into the welding holder. 4.6 The welding machine is tested with trial runs to verify correct setup and functionality.
5. Adjust welding parameters for each welding position	4.1 Welding positions and related welding specifications are understood. 4.2 **Welding parameters** such as current, voltage, travel

	speed, and polarity are identified according to the welding position. 4.3 Welding machine parameters are adjusted of each specific welding position. 4.4 Trial welds are conducted to ensure parameters produce acceptable weld quality in each position.

Range of Variables

Variable	**Range** (Includes but not limited to):
1. Welding procedure specifications (WPS)	1.1 Joint information 1.2 Technical information 1.3 Joint preparation 1.4 Welding details 1.5 Back gouging
2. Standards	2.1 AWS D1.1 2.2 ASME IX 2.3 ISO 5018
3. Base materials	3.1 **Steel (for 1G, 2G and 3G)** 3.1.1 Mild steel 3.1.2 Carbon steel 3.1.3 Alloy steels
4. Welding tools	4.1 Electrodes based on the size of job/base materials 4.2 Clamps 4.3 Grinders 4.4 Chipping hammers 4.5 Safety gear
5. PPE	5.1 Safety glasses 5.2 Ear plugs 5.3 Gloves 5.4 Apron 5.5 Helmet 5.6 Mask 5.7 Safety shoes
6. Welding joint types	6.1 Butt joints 6.2 Lap joints 6.3 Corner joints 6.4 T-joints 6.5 Thickness of parts 6.6 Root opening 6.7 Root face 6.8 Bevel angle 6.9 Groove angle 6.10 Effective throat thickness
7. Welding parameters	7.1 Current 7.2 Voltage

	7.3 Travel speed 7.4 Polarity 7.5 Number of passes

Curricular Content Guide

1. Underpinning Knowledge	1.1. Welding Procedure Specifications (WPS) and standards. 1.2. Identify base materials and material type and welding position 1.3. Identify necessary welding tools. 1.4. Identify welding joint types, dimensions, angles, and fit-up requirements. 1.5. Inspect the welding machine for safety, damage, or wear. 1.6. Identify and adjust parameters for each position.
2. Underpinning Skills	2.1 Identifying base materials 2.2 Selecting welding consumables based on base materials and welding position 2.3 Inspecting welding consumables for defects 2.4 Identifying required welding tools 2.5 Inspecting welding tools to ensure cleanliness, functionality, and freedom from defects 2.6 Arranging welding tools systematically for easy accessibility 2.7 Setting welding parameters according to welding procedure specifications 2.8 Installing and feeding electrodes correctly into welding holders 2.9 Testing welding machines with trial runs to verify setup and functionality 2.10 Storing tools securely to maintain safety and prevent damage
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4 Resource Implications	4.1 Workplace (simulated or actual) 4.2 Personal protective equipment (PPE) 4.3 Tools and equipment 4.4 Materials 4.5 Projector

	4.6 Stationary 4.7 Learning manual

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 selected base materials and consumables 1.2 identified welding joints 1.3 set up welding machines 1.4 adjusted welding parameters for each welding positions
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: **CARRY OUT SHIELDED METAL ARC WELDING**	**Nominal Duration:** 80 Hrs**.**	**Unit Code:** SICIP-LE-WEL-03-O
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to carry out shielded metal arc welding. It specifically includes the tasks of identifying and preparing work requirements, selecting welding job, equipment and job holding devices, performing welding in different positions, and cleaning/maintaining the workplace.		

Elements and Performance Criteria:
(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify and prepare work requirements	1.1 Drawings are interpreted to shielded metal arc welding conforming to the specifications. 1.2 **Welding machine**, **Tools and equipment** are selected according to the requirements of the welding works. 1.3 **Base metal/plates** and **electrodes** are selected according to requirements of the job. 1.4 **PPE** is selected and used 1.5 Base metals are prepared as per requirement. 1.6 Welding machine operation are demonstrated as per requirement. 1.7 **Welding joint, position and process** are demonstrated according to the job requirement.
2. Select welding job, equipment and job holding devices	2.1 Welding equipment and holding devices are set up and adjusted in accordance with the job requirements. 2.2 Welding area guards, work table/floor, dust collection devices are checked according to worksite procedure. 2.3 Welding machine is prepared and bead practices indifferent positions are carried out.
3. Perform welding in different positions	3.1 Welding machine performance is checked conforming to the job requirement. 3.2 Welding in different positions are carried out. 3.3 Welds are cleaned, checked for quality and defects are identified. 3.4 Defects are rectified to meet the standards of job specifications.
4. Clean/maintain the workplace	4.1 Workplace is inspected to identify any potential hazards. 4.2 Cleaning tasks are performed according to the cleanliness protocols. 4.3 Work areas are organized, and tools and equipment are stored.

	4.4 Equipment and machinery are cleaned and maintained. 4.5 The workplace is kept free from debris and obstructions to ensure a safe and efficient working environment.

Range of Variables

Variable	**Range** (Includes but not limited to):
1. Welding Machine	1.1 AC welding machine (dry/wet). 1.2 DC welding machine.
2. Tools and equipment	2.1 Clamps 2.2 Chipping hammer. 2.3 Pliers. 2.4 Wire brush 2.5 Weld gauge 2.6 Welding table with positioner 2.7 Job holding devices/fixture 2.8 Hand grinder 2.9 Hand drill
3. Base metal plates	3.1 MS Plates 3 mm to 12 mm thickness
4. Electrodes	4.1 8,10 and 12 SWG (4, 3.2 and 2.5 mm)
5. Welding joint and position	5.1 F –means fillet. 5.2 G –means groove 5.3 These two welds are 4 (four) in position. 5.4 Pos. No. – 1 is flat position. 5.5 Pos. No. – 2 is Horizontal position. 5.6 Pos. No. – 3 is Vertical position. 5.7 Pos. No. – 4 is Overhead position. 5.8 Example- 1F indicates the flat position of fillet welding 5.9 1G indicates the flat position of groove welding and so on.
6. PPE	6.1 Protective musk. 6.2 Eye shield. 6.3 Goggles. 6.4 Safety shoes. 6.5 Apron. 6.6 Helmet. 6.7 Leather gloves. 6.8 Full sleeve leather jacket

Curricular Content Guide

1. Underpinning Knowledge	1.1 Methods of interpreting Drawings of shielded metal arc welding 1.2 Selection of Welding machine, Tools and equipment 1.3 Selection of Base metal/ plates and electrodes 1.4 PPE and their uses 1.5 Means of preparing Base metals

	1.6 Welding machine operation 1.7 position of Welding joints 1.8 Safe work practices when performing welding
2. Underpinning Skills	2.1 Checking welding machine performance in conformance to the job requirement. 2.2 Performing routine maintenance 2.3 Preparing the welding machine 2.4 Performing requirement of the welding job 2.5 Marking, cutting and setting welding job 2.6 Setting up welding equipment and holding devices 2.7 Adjusting welding equipment and holding devices in accordance with the job requirements. 2.8 Performing welding as per the job requirement in accordance with work positions for 1F, 2F, 3F, 4F and 1G, 2G, 3G, 4G. 2.9 Cleaning and checking welds for quality and identifying defects 2.10 Rectifying defects to meet the Welding Procedure Specification (WPS). 2.11 Checking welding area guards, work table/floor, dust collection devices in accordance to worksite procedure.
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4 Resource Implications	4.1 Workplace (simulated or actual) 4.2 Personal protective equipment (PPE) 4.3 Tools and equipment 4.4 Materials 4.5 Projector 4.6 Stationery 4.7 Learning manual

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 identified and prepared work requirements 1.2 selected welding job, equipment and job holding devices 1.3 performed welding 1.4 cleaned/maintained the workplace

2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: **PERFORM GAS WELDING**	**Nominal Duration:** 60 Hrs.	**Unit Code:** SICIP-LE-WEL-04-O
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to perform gas welding. It specifically includes the task of preparing for gas welding, carry out gas welding, carry out gas cutting and cleaning and storing tools and equipment.		

Elements and Performance Criteria:
(Terms in the performance criteria that are written in **<u>bold and underlined</u>** are elaborated in the range of variables).

Elements of Competency	**Performance Criteria**
1. Prepare for gas welding	1.1 PPE is selected & used. 1.2 **<u>Tools, welding equipment</u>** and **<u>materials</u>** are selected and collected. 1.3 Gas welding sets and accessories are gathered and set up according to the job specifications. 1.4 Drawings are interpreted to produce component to the job specifications. 1.5 Jig, fixture and positioner are set up for gas welding.
2. Carry out gas welding	2.1 Gas welding torch is adjusted to perform different types of flame for welding. 2.2 Gas welding set performance is checked conforming to the job requirement. 2.3 Gas welding is performed in different **<u>welding joint and position</u>** 2.4 Welds are cleaned, checked for quality and defects are identified. 2.5 Defects are rectified to meet the standards of job specifications.
3. Perform gas cutting	3.1 Gas cutting torch is set for cutting materials as per job thickness. 3.2 Gas cutting set performance is checked conforming to the job requirement. 3.3 **<u>Gas cutting</u>** is performed as per the job requirement. 3.4 Cut pieces rough edges are removed, cleaned, checked for quality. 3.5 Defects are identified and corrective action is taken according to standard cutting procedures.
4. Clean and store tools and equipment	4.1 Tools, equipment and cutting torch are cleaned. 4.2 Work place is cleaned 4.3 Waste materials are disposed in proper place. 4.4 Tools, equipment and finished job are stored safely in appropriate location according to standard place and procedures.

Range of Variables

Variable	**Range** (Includes but not limited to):
1. Tools	1.1 C-clamp 1.2 Chipping hammer 1.3 Ball pin hammer 1.4 Center punch 1.5 Hack saw 1.6 File 1.7 Chisel 1.8 Pliers 1.9 Wire brush 1.10 Weld gauge 1.11 Blacksmith tong 1.12 Welding table with positioner 1.13 Job holding devices/fixture 1.14 Hand grinder 1.15 Hand drill 1.16 Soldering tools
2. Welding equipment	2.1 Gas welding set and its accessories 2.2 Oxygen cylinder 2.3 Acetylene cylinder 2.4 Cylinder pressure gauge (oxy/acetylene) 2.5 Working pressure gauge (oxy/acetylene) 2.6 Oxygen regulator 2.7 Oxygen hose 2.8 Spark lighter 2.9 Apparatus wrench 2.10 Welding torch 2.11 Torch tip/ nozzle 2.12 Oxygen hose connections 2.13 Acetylene hose connections 2.14 Acetylene regulator 2.15 Acetylene hose 2.16 Twin hose 2.17 Welder hose joints 2.18 Coupler joint socket 2.19 Coupler joint plug 2.20 Cutting torch
3. Materials	3.1 Mild steel 3.2 Carbon steel 3.3 Cast iron 3.4 Aluminum 3.5 Brass 3.6 Brazing materials 3.7 Soldering materials
4. Welding joint and	4.1 Joints – Lap, butt, Tee and corner.

position	4.2 Position –Flat and Horizontal
5. Gas cutting	5.1 Straight cutting 5.2 Zig zag cutting 5.3 L cutting 5.4 Round cutting

Curricular Content Guide

1. Underpinning Knowledge	1.1 Understand the importance of selecting the correct personal protective equipment (PPE). 1.2 Recognize the process of gathering and setting up the gas welding set. 1.3 Importance of setting up jigs, fixtures, and positioners. 1.4 Recognize the need to adjust the gas welding torch. 1.5 Importance of checking the performance of the gas welding set. 1.6 Procedure for performing gas welding in various joints and positions. 1.7 Cleaning process of welds and how to check for defects or quality issues. 1.8 Process of checking the gas cutting set. 1.9 Process of performing gas cutting based on the job requirements. 1.10 Importance of removing rough edges from cut pieces and checking them for quality. 1.11 Recognize the importance of keeping the workplace clean and free from debris.
2. Underpinning Skills	2.1 Selecting and wearing appropriate PPE (Personal Protective Equipment). 2.2 Selecting and collecting the necessary tools, welding equipment, and materials. 2.3 Adjusting the gas welding torch to produce different types of flame. 2.4 Checking the performance of the gas welding set. 2.5 Performing gas welding on various joints and in different positions. 2.6 Cleaning welds and inspect them for quality, identifying any defects or imperfections. 2.7 Adjusting the gas cutting torch to the correct settings based on the material thickness for accurate cutting. 2.8 Performing gas cutting according to the job requirements, ensuring precision and quality. 2.9 Cleaning tools, equipment, and the cutting torch after use.
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns

	3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4 Resource Implications	4.1 Workplace (simulated or actual) 4.2 Personal protective equipment (PPE) 4.3 Tools and equipment 4.4 Gas welding machine 4.5 Gas cutting machine 4.6 Materials

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 prepared for gas welding 1.2 performed gas welding 1.3 performed gas cutting 1.4 cleaned and stored tools and equipment
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: **CARRY OUT GAS METAL ARC WELDING**	**Nominal Duration:** 60 Hrs.	**Unit Code:** SICIP-LE-WEL-05-O
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to carry out gas metal arc welding. It specifically includes the tasks of identifying and preparing work requirements, selecting welding job, equipment and job holding devices, performing GMAW or MIG welding job and cleaning/maintaining the workplace.		

Elements and Performance Criteria:
(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify and prepare work requirements	1.1 GMAW or MIG welding drawings are interpreted. 1.2 MIG welding machine, **<u>Tools and equipment</u>** are selected according to the requirements. 1.3 **<u>PPEs</u>** are selected and used. 1.4 Safe work practices are observed and personal protective equipment (PPE) are worn as required. 1.5 **<u>Base metal/ plates</u>**, **<u>wire electrode sizes</u>** and **<u>shielding gas</u>** are selected according to requirements of the job. 1.6 Base metals and **<u>GMAW weld area</u>** are prepared as per requirement. 1.7 MIG welding machine operations is demonstrated 1.8 **<u>Welding joint and position</u>** are demonstrated according to the job requirement.
2. Select welding job, equipment and job holding devices	2.1 Welding job, equipment and holding devices are identified as per requirement. 2.2 Welding machine is prepared as per job requirement. 2.3 Welding equipment and holding devices are set up and adjusted in accordance with the job requirements
3. Perform GMAW or MIG welding	3.1 MIG welding machine and welding torch performance is checked in accordance with the job requirement. 3.2 Amperage and gas flow is set in accordance with work piece plate thickness. 3.3 Welding is performed in Butt and Tee joints in flat and horizontal positions 3.4 Welds are cleaned, checked for **<u>quality test</u>** and **<u>defects</u>** are identified. 3.5 Defects are rectified to meet the standards of job specifications.
4. Clean/maintain the workplace	4.1 Workplace is inspected to identify any potential hazards. 4.2 Cleaning tasks are performed according to the cleanliness protocols. 4.3 Work areas are organized, and tools and equipment are

	stored. 4.4 Equipment and machinery are cleaned and maintained. 4.5 The workplace is kept free from debris and obstructions to ensure a safe and efficient working environment.

Range of Variables

Variable	**Range** (Includes but not limited to):
1. Tools and equipment	1.1 Clamps 1.2 Chipping hammer 1.3 Pliers 1.4 Wire brush 1.5 Weld gauge 1.6 Welding table with positioner 1.7 Job holding devices/fixture 1.8 Hand grinder 1.9 Hand drill 1.10 MIG torch 1.11 Gas regulators 1.12 MIG wire and consumables 1.13 Torch tips
2. PPE	2.1 Protective musk 2.2 Dark eye lenses 2.3 Goggles 2.4 Safety shoes 2.5 Protective clothing. 2.6 Apron 2.7 Helmet 2.8 Leather gloves 2.9 Full sleeve leather jacket
3. Base metal/ plates	3.1 MS Plate 3.2 Carbon steel. 3.3 Stainless steel. 3.4 Aluminum
4. Wire sizes	4.1 Wire diameter- 0.7 – 2.4 mm
5. Shielding gas	5.1 Inert gas- Argon, helium, 5.2 Reactive gases – oxygen or carbon dioxide.
6. GTAW weld area	6.1 Direction of travel. 6.2 Shielding gas. 6.3 Contact tube. 6.4 Wire electrode (consumable). 6.5 Molten metal. 6.6 Solidified weld metal.

	6.7 Work piece. 6.8 Electric arc. 6.9 Copper shoe.
7. Welding joint and position	7.1 Butt joint and Tee joint. 7.2 F, 2F and 1G, 2G
8. PPE	8.1 Protective musk 8.2 Dark eye lenses 8.3 Goggles 8.4 Safety shoes 8.5 Protective clothing. I. Apron II. Helmet III. Leather gloves 8.6 Full sleeve leather jacket
9. Quality test	9.1 **Destructive test:** 9.1.1 Tensile test 9.1.2 Bend test 9.1.3 Hardness test 9.1.4 Break test 9.2 **Non – Destructive test:** 9.2.1 Radiographic test 9.2.2 Ultrasonic test 9.2.3 Magnetic particle test 9.2.4 Leak test 9.3 Visual test
10. Defects	10.1 Lack of penetration. 10.2 Excess of penetration. 10.3 Porosity. 10.4 Crack. 10.5 Slag Inclusion. 10.6 Distortion 10.7 Undercut. 10.8 Lack of fusion 10.9 Notches. 10.10 Irregular shape. 10.11 Dimension. 10.12 Poor weld start. 10.13 Wire stubbing. 10.14 Excessive spatter. 10.15 Bum through. 10.16 Convex bead.

Curricular Content Guide

Welding (1G, 2G and 3G)

1. Underpinning Knowledge	1.1 Operation and application of GMAW or MIG welding processes 1.2 Selection procedure of MIG welding machine 1.3 Welding tools and equipment and their use 1.4 Types of base metal/ plates 1.5 Wire size and their application 1.6 Selection procedure for shielding gas 1.7 Operation of a MIG welding machine 1.8 Types of welding joints and positions 1.9 Safe working practices in welding jobs
2. Underpinning Skills	2.1 Checking MIG welding machine and welding torch performance 2.2 Setting amperage and gas flow in relation with the welds plate thickness 2.3 Performing welding in different joint (Butt and Tee) and position (Flat and Horizontal) 2.4 Cleaning and checking welds for quality test and defects 2.5 Rectifying defects to meet the standards of job specifications. 2.6 Storing Tools, equipment and finished products safely in appropriate location according to standard place and procedures.
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	4.1 Workplace (simulated or actual) 4.2 Personal protective equipment (PPE) 4.3 Tools and equipment 4.4 MIG welding machine 4.5 Materials

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 identified and prepared work requirements 1.2 selected welding job, equipment and job holding devices 1.3 performed GMAW or MIG welding 1.4 cleaned/maintained the workplace
2. Methods of Assessment	Competency should be assessed by: 2.1 Written examination 2.2 Practical demonstration 2.3 Oral questioning

	2.4 Portfolio review
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: **CARRY OUT GAS TUNGSTEN ARC WELDING**	**Nominal Duration:** 60 Hrs**.**	**Unit Code:** SICIP-LE-WEL-06-O
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to carry out gas tungsten arc welding. It specifically includes the tasks of identifying and preparing work requirements, selecting welding job, equipment & job holding devices, performing GTAW or TIG welding job and cleaning/maintaining the workplace.		

Elements and Performance Criteria:
(Terms in the performance criteria that are written in **<u>bold and underlined</u>** are elaborated in the range of variables).

Elements of Competency	**Performance Criteria**
1. Identify and prepare work requirements	1.1 Gas tungsten arc welding drawings are interpreted and confirm specifications 1.2 **<u>PPEs</u>** are selected and used 1.3 TIG welding machine, **<u>Tools and equipment</u>** are selected according to the requirements. 1.4 **<u>Base metals/ plate</u>**, **<u>filler metal</u>,** Tungsten **<u>electrodes</u>** and **<u>shielding gas</u>** are selected according to requirements of the job. 1.5 Base metals and **<u>GTAW welds area</u>** are prepared as per requirement. 1.6 **<u>Welding joint and position</u>** are demonstrated according to the job requirement.
2. Select welding job, equipment and job holding devices	2.1 Welding job, equipment and holding devices are identified as per requirement. 2.2 Welding machine is prepared as per job requirement. 2.3 Welding equipment and holding devices are set up and adjusted in accordance with the job requirements.
3. Perform GTAW or TIG welding	3.1 TIG welding machine and welding torch performance is checked conforming to the job requirement. 3.2 Amperage and gas flow is set according to the welds plate thickness and gas flow cup sizes. 3.3 Welding is performed as per the job requirement and welds in different joint and position. 3.4 Welds are cleaned, checked for **<u>quality test</u>** and **<u>defects</u>** are identified. 3.5 Defects are rectified to meet the standards of job specifications.
4. Clean/maintain the workplace	4.1 Workplace is inspected to identify any potential hazards. 4.2 Cleaning tasks are performed according to the cleanliness protocols. 4.3 Work areas are organized, and tools and equipment

	are stored. 4.4 Equipment and machinery are cleaned and maintained. 4.5 The workplace is kept free from debris and obstructions to ensure a safe and efficient working environment.

Range of Variables

Variable	**Range** (Includes but not limited to):
1. PPE	1.1 Protective mask 1.2 Dark eye lenses 1.3 Goggles 1.4 Safety shoes 1.5 Protective clothing 1.6 Apron 1.7 Helmet 1.8 Leather gloves 1.9 Full sleeve leather jacket
2. Tools and equipment	2.1 Clamps 2.2 Chipping hammer 2.3 Pliers 2.4 Wire brush 2.5 Weld gauge 2.6 Welding table with positioner 2.7 Job holding devices/fixture 2.8 Hand grinder 2.9 Hand drill 2.10 GTAW torch. 2.11 Electrodes (Tungsten). 2.12 Cups. 2.13 Collects. 2.14 Gas diffusers. 2.15 GTAW power suppliers
3. Base metal/ plates	3.1 MS Plates 3.2 Carbon steel 3.3 Stainless steel 3.4 Aluminum
4. Filler metal	4.1 Base metal material 4.2 Stainless steel 4.3 Carbon steel 4.4 Aluminum
5. Electrodes	5.1 Tungsten 5.2 Tungsten alloy
6. Shielding gas	6.1 Inert gas- Argon, helium 6.2 Reactive gases – nitrogen, oxygen or carbon dioxide. 6.3 Mixtures of inert and reactive gases
7. GTAW weld area	7.1 GTAW head

	7.2 Power 7.3 Shielding gas 7.4 Contact tube 7.5 Tungsten electrode (non-consumable). 7.6 Weld bead 7.7 Direction of weld 7.8 Filler rod 7.9 Electric arc 7.10 Copper shoe (optional)
8. Welding joint and position	8.1 Lap joint 8.2 butt joint 8.3 Tee joint 8.4 1F, 2F and 2G, 3G
9. Quality test	9.1 **Destructive test** 9.1.1 Tensile test 9.1.2 Bend test 9.1.3 Hardness test 9.1.4 Break test 9.2 **Non-Destructive test:** 9.2.1 Radiographic test 9.2.2 Ultrasonic test 9.2.3 Magnetic particle test 9.2.4 Leak test 9.3 Visual test
10. Defects	10.1 Lack of penetration 10.2 Excess of penetration 10.3 Porosity 10.4 Crack 10.5 Slag Inclusion 10.6 Distortion 10.7 Undercut 10.8 Lack of fusion 10.9 Notches. 10.10 Irregular shape 10.11 Dimension.

Curricular Content Guide

1. Underpinning Knowledge	1.1 Gas Tungsten Arc Welding Drawings interpretation 1.2 TIG welding machine, Tools and equipment 1.3 PPE requirements when performing TIG welding 1.4 Base metals plate, filler metal, Tungsten electrodes and shielding gas 1.5 Types of Base metals 1.6 GTAW welds area preparation 1.7 operation of TIG Welding machine 1.8 Types of Welding joint and position 1.9 Safe work practices

2. Underpinning Skills	2.1 Checking TIG welding machine and welding torch performance 2.2 Setting amperage and gas flow 2.3 Marking, cutting and setting welds as per the requirement. 2.4 Setting up of welding equipment and holding devices 2.5 Performing welding as per the job requirement 2.6 Welding in 1F, 2F and 2G, 3G. 2.7 Cleaning and checking welds for quality test 2.8 Identifying defects 2.9 Rectifying defects to meet the Welding Procedure specification (WPS).
3. Underpinning Attitudes	3.1. Commitment to occupational health and safety 3.2. Promptness in carrying out activities 3.3. Sincere and honest to duties 3.4. Environmental concerns 3.5. Eagerness to learn 3.6. Tidiness and timeliness 3.7. Respect for rights of peers and seniors in the workplace 3.8. Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	4.1 Workplace (simulated or actual) 4.2 Standard operating procedure 4.3 Workplace documents, signs and symbols 4.4 Codes of conduct 4.5 Projector 4.6 Learning manual 4.7 Tools, equipment and facilities appropriate to the process or activities. 4.8 Materials are relevant to the proposed activity.

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 identified and prepared work requirements 1.2 selected worlding job, equipment and job holding devices 1.3 performed GTAW or TIG welding 1.4 cleaned/maintained the workplace
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated

	work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

End of the Competency Standard

Workshop/Lab Facility Standard

Course Name:	**Welding (1G, 2G & 3G)**
Number of Trainees:	25

Course-wise Training Space (Theoretical Classroom, Workshop/ Lab/ Classroom cum Workshop):

- Classroom – 350 sft (33 sqm)
- Workshop/ lab – 800 sft (75 sq) OR
- Classroom cum workshop – 1000 sft (93 sqm)

Major Training Equipment and Training Facilities:

Sl. No.	Major Equipment and Training facilities	Required facilities
1.	Computers/ Laptop	01
2.	Multimedia Projector	01
3.	Internet connectivity	01
4.	AC/DC Arc welding machine	10
5.	Gas welding set	03
6.	MIG welding machine	05
7.	TIG welding machine	05
8.	Gas cutting set	02
9.	Spare cylinder set (Argon, oxygen and acetylene)	01
10.	Power grinding machine	01
11.	Hand grinding machine	04
12.	Bench vice	10
13.	Tool box set for welding workshop	10
14.	Fire extinguisher	04
15.	Welding booth with exhaust system	10
16.	First aid box	03

The following conditions must be fulfilled –

- The institute shall not use the same facilities for any other projects/organizations offering a similar course.
- The institute must provide sufficient evidence to prove ownership of the proposed training equipment.

The list denotes the minimum training equipment and facility required to effectively conduct training for a specific course. Additionally, the institute must ensure that all other necessary training tools, equipment, and furniture are available to meet the requirement of competency standards (CS) provided by SICIP.

For the operation of training course on Welding (1G, 2G & 3G) the institute must ensure the availability of at least 80% of the major training equipment and training facilities (according to the CS) to be eligible for SICIP training delivery. If the score is below 80%, the remaining equipment and facilities need to be installed before the commencement of the training.

The institute will also provide all other hand tools and power tools as per CS for 25 trainees. Also, they will arrange adequate seating arrangement and classroom setup for the 25 trainees.